

Project Title: "LuminaEdit Core"

1. Document Management

- **Requirement:** The user should be able to open a file from the disk into a "Buffer."
- **Requirement:** The user can work on multiple files at the same time (switching between tabs/buffers).
- **Requirement:** The user must be able to save the current state of a buffer back to the file system.

2. Text Manipulation (The Core Logic)

- **Requirement:** The user can insert a character at the current cursor position.
- **Requirement:** The user can delete a character (backspace) or a whole line.
- **Requirement:** The editor must handle very large files efficiently. If I insert a character at the start of a 1-million-line file, the editor should not freeze.
- **Requirement:** The system should support "Bulk Replace" (find all instances of "old" and change to "new").

3. Navigation & Cursor Tracking

- **Requirement:** The system must track the cursor's

(x, y)

coordinate (line number and column number).

- **Requirement:** The user can move the cursor up, down, left, and right.
- **Requirement:** The cursor should automatically wrap to the next line if it goes past the end of the current line during movement.

4. Search & Analytics

- **Requirement:** The user can search for a word. The system should return all line numbers where the word appears.
- **Requirement:** The user wants a "Code Statistics" feature that shows:
 - The total number of characters.
 - The longest line in the file.

- The frequency of specific keywords (like int, if, while).

5. Undo/Redo System (The History)

- **Requirement:** Every action (insertion, deletion) must be reversible.
- **Requirement:** The user should be able to "Undo" their last 10-20 actions and "Redo" them if they change their mind.

6. Extension/Plugin System

- **Requirement:** The editor should allow for "Formatters" or "Analyzers" to be plugged in.
 - Example: A "C++ Formatter" that adds a space after if.
 - Example: A "Word Counter" that alerts the user when they exceed 1000 words.
- **Note:** You must be able to add a new formatter without changing the existing editor code (**Open/Closed Principle**).